

Multi-Disease Prediction System

Leveraging Hybrid Machine Learning for Predictive Healthcare

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Introduction & Motivation

Automating Disease Prediction: A Hybrid ML Approach



The Growing Need

The healthcare sector urgently requires automated decision-support systems. These systems must swiftly analyze both subjective symptoms and structured clinical data to facilitate rapid disease identification and improve patient outcomes.



Our Primary Objective

To engineer a comprehensive multi-disease prediction system, leveraging two distinct yet complementary machine learning architectures for enhanced diagnostic accuracy and utility.



System Overview & Architecture

The Hybrid Approach: Bridging Symptom-Based and Clinical Data

Architecture 1: Symptom-Based Prediction

This architecture focuses on predicting one of 42 common diseases based on patient-reported symptoms.

- **Input:** Textual symptoms provided by the patient.
- **Preprocessing:** Extensive one-hot encoding and sparse matrix transformation to handle the diverse symptom input.
- **Model:** Utilises a highly efficient XGBoost Classifier.
- **Output:** Predicts a single disease from a list of forty-two potential conditions.

Architecture 2: Structured Clinical Pipeline

Designed for robust predictions using structured clinical datasets, such as those for Diabetes or Heart Disease.

- **Input:** Measurable physiological attributes (e.g., blood glucose levels, BMI, blood pressure).
- **Pipeline:** Incorporates sophisticated data preparation, feature engineering.
- **Model:** An advanced Ensemble Stacking Classifier with a Logistic Regression meta-learner for superior performance.

Deployment Goal: Both models are seamlessly deployed via an intuitive Streamlit application, offering real-time predictions for healthcare professionals.

Sources and Transformations for Accurate Predictions

Symptom Model Data

Source: dataset.csv (featuring 42 distinct disease classes).

Features: Comprises up to 17 symptom columns, which are transformed into a 133-feature one-hot encoded sparse matrix to capture all possible symptom combinations.

Supporting Data: symptom_description.csv and symptom_precaution.csv are utilised to enhance the interpretability of predictions, providing context and guidance.

Structured ML Pipeline Data

Source: pima_diabetes.csv from the UCI Repository, a well-established dataset for diabetes prediction.

Features: Consists of 8 numerical clinical features, including vital metrics like glucose levels, BMI, and blood pressure.

Key Preprocessing: Involves robust missing value handling, StandardScaler application on numerical features for normalisation, and cross-validation-safe Target Encoding to prevent data leakage.

Data Split: An 80-20 train-test split is employed to ensure rigorous model evaluation.

XGBoost: Powering Predictions on Sparse Symptom Data

Our symptom-based prediction model leverages the power of XGBoost to accurately identify diseases from patient-reported symptoms.

01

User Symptom Input

The process begins with patients entering their symptoms, which form the initial data points for prediction.

02

One-Hot Encoding

These symptoms are then transformed into a high-dimensional binary vector using one-hot encoding, creating a sparse matrix suitable for machine learning.

03

XGBoost Classifier

The encoded data is fed into the XGBoost Classifier. This model was chosen for its exceptional ability to handle sparse, high-dimensional input and effectively capture complex, non-linear relationships between symptoms and diseases.

04

Disease Prediction

The XGBoost model then outputs a prediction, identifying one of the 42 common diseases with high accuracy.

Methodology - Structured Pipeline Model Selection

Ensemble Stacking for Clinical Data Precision

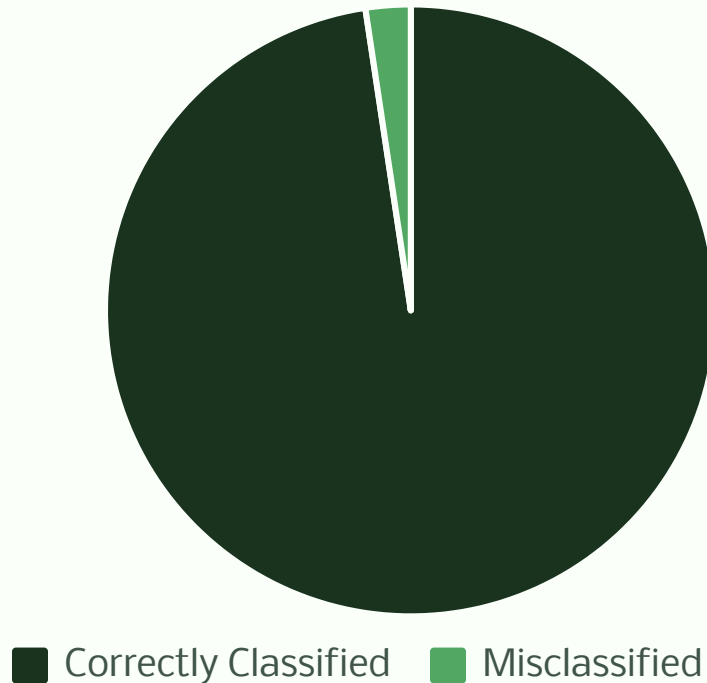
For structured clinical datasets, we employ a robust methodology involving comprehensive model selection and an advanced stacking ensemble.



Model Evaluation

The `models.py` script rigorously evaluates a diverse range of algorithms, including Logistic Regression, Random Forest, CatBoost, and SVM, utilising cross-validation to assess their individual performance.

High Accuracy in Multi-Disease Prediction



The Symptom-Based Model, powered by the XGBoost Classifier, demonstrated exceptional performance in multi-disease prediction.

- **Test Accuracy:** An impressive 97.62% was achieved on the test dataset.
- **F1-Score (Weighted):** A robust F1-Score of 0.98, indicating excellent precision and recall across all disease classes.
- **Key Observation:** Out of 42 possible diseases, only 1 case was misclassified, specifically "Paralysis," highlighting the model's strong discriminative power.



Conclusion & Future Work

Scalable and Interpretable Hybrid System for Medical Decision Support



Successful Development

We have successfully engineered and deployed a scalable, interpretable hybrid system offering both breadth (42 diseases via symptoms) and depth (clinical pipeline for specific conditions).



Future Expansion

Our next steps involve expanding the symptom dataset with greater diversity and acquiring real-world clinical data to enhance model robustness and generalisability.



Integration and Accessibility

We plan to develop a mobile-friendly interface and integrate APIs to connect with hospital databases, enabling seamless access and real-time utility for healthcare providers.

Thank You

We appreciate your time and welcome any questions you may have regarding our Multi-Disease Prediction System.

